



Lecture 6

6.1 Convolution

畳み込み

6.1 Convolution

Convolution of Functions

We define the **convolution** of two (continuous-time) functions f and g by

$$(f * g)(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x-t)g(t)dt \quad (6.1)$$

Convolution

Our Textbook

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-t)g(t)dt$$

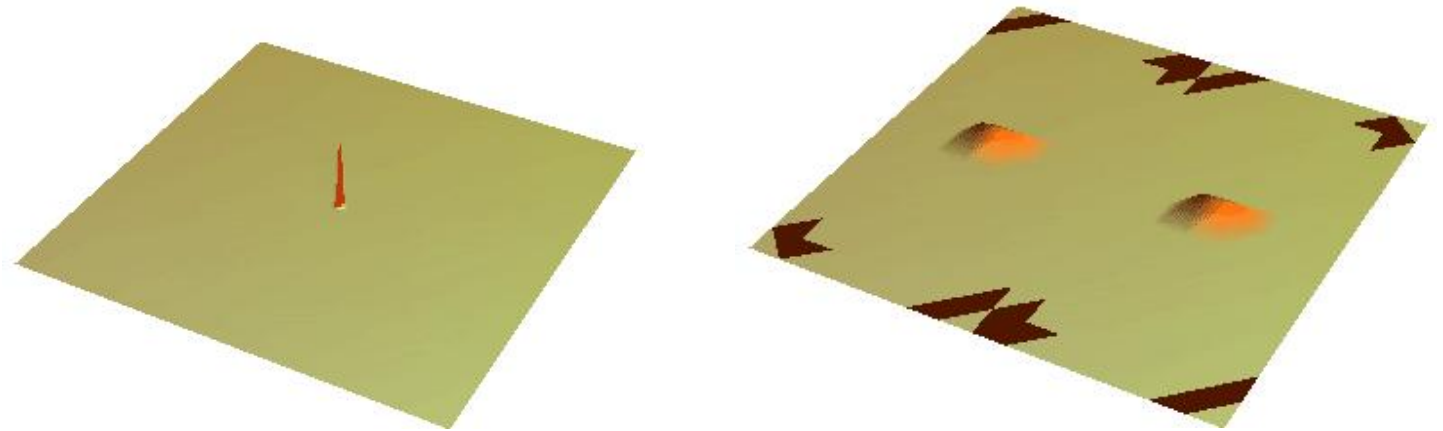
Convolution

Wikipedia <https://en.wikipedia.org/wiki/Convolution>

6.1 Convolution

What is the insight of **Convolution**?

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x - t)g(t)dt$$



6.1 Convolution

Review *Calculus I*

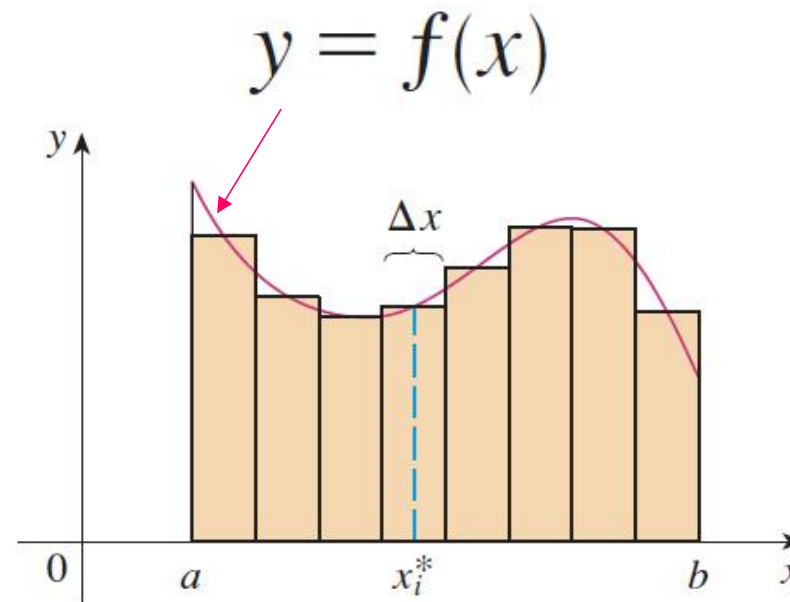
One-Variable Calculus – **Definite integral (定積分)** of f

Riemann Sum

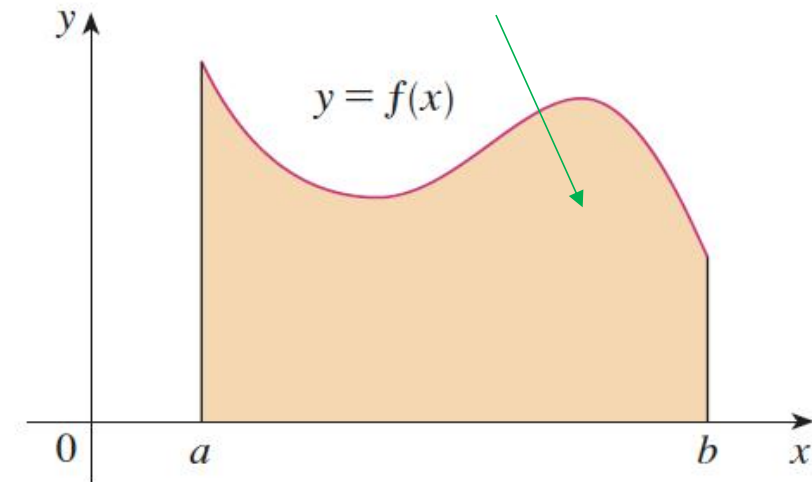
$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \lim_{n \rightarrow \infty} [f(x_1^*) \Delta x + f(x_2^*) \Delta x + \dots + f(x_n^*) \Delta x]$$

One-Variable Function

$$y = f(x)$$



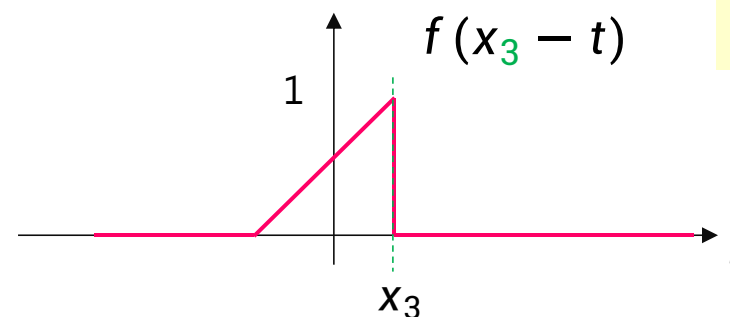
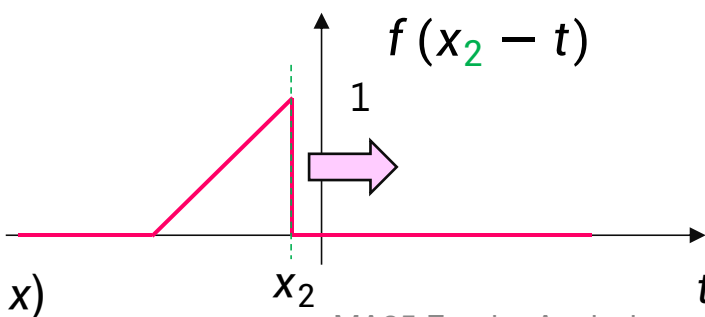
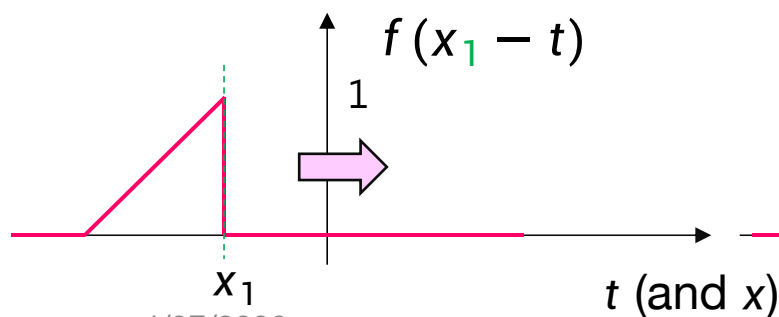
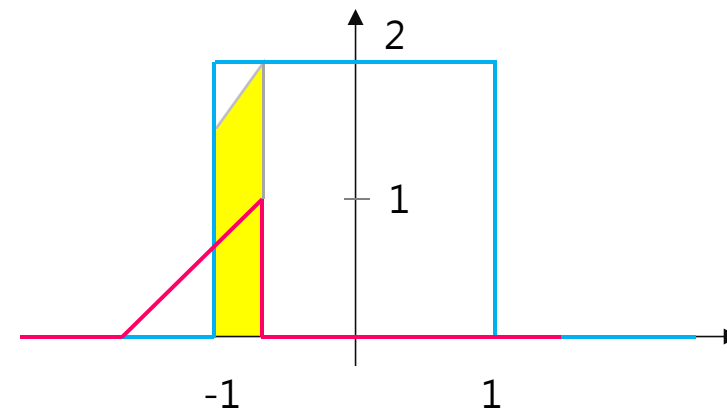
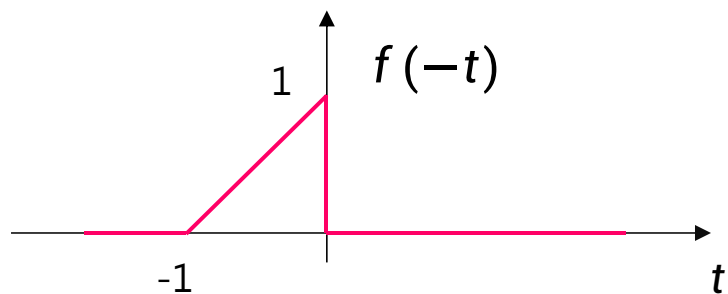
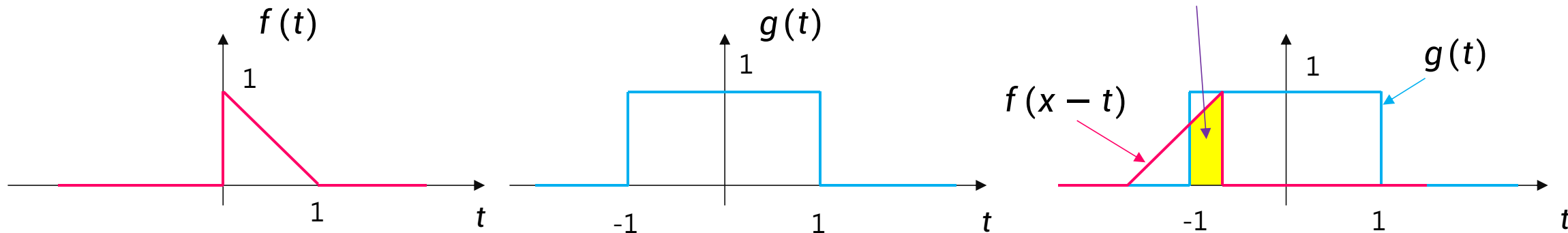
$$\text{Area (面積)} = \int_a^b f(x) dx$$



6.1 Convolution

Case 1

$$h = (f * g)(x) = \int_{-\infty}^{\infty} f(x - t)g(t)dt$$

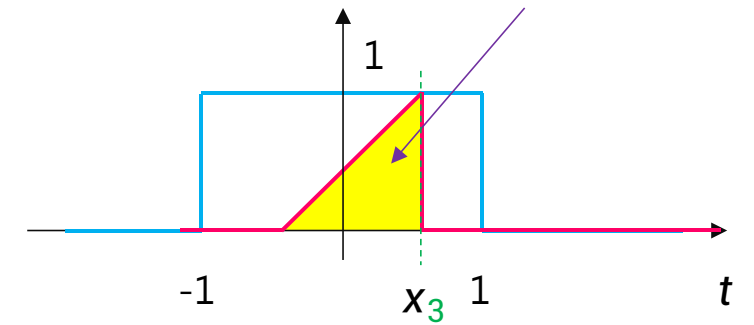
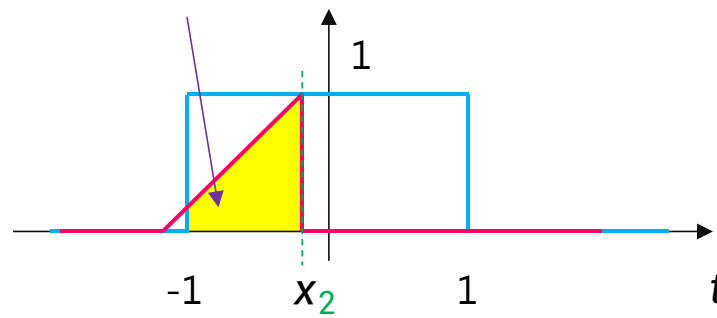
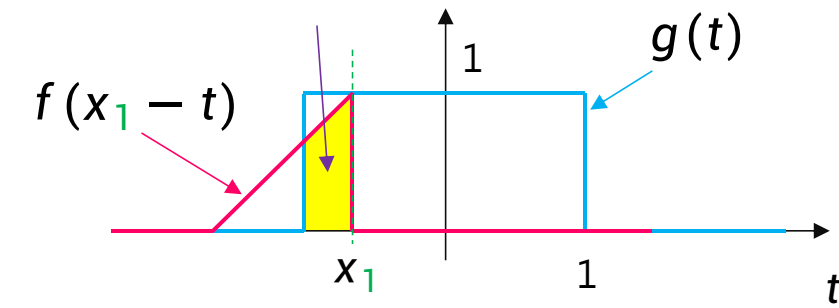


$$x_1 < x_2 < x_3$$

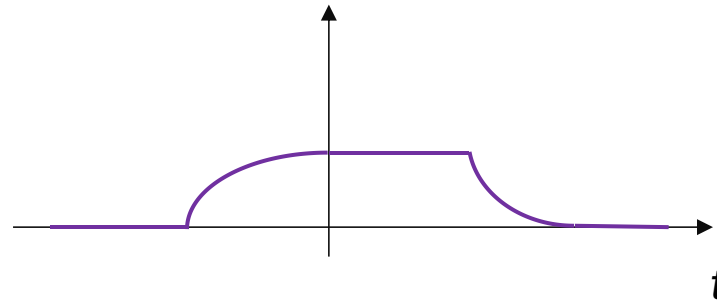
6.1 Convolution

Case 1

$$(f * g)(x_1) = \int_{-\infty}^{\infty} f(x_1 - t)g(t)dt \quad (f * g)(x_2) = \int_{-\infty}^{\infty} f(x_2 - t)g(t)dt \quad (f * g)(x_3) = \int_{-\infty}^{\infty} f(x_3 - t)g(t)dt$$



$$h = (f * g)(x) = \int_{-\infty}^{\infty} f(x - t)g(t)dt$$

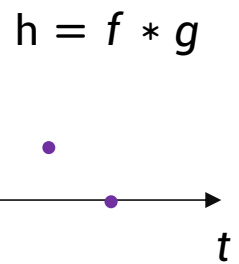
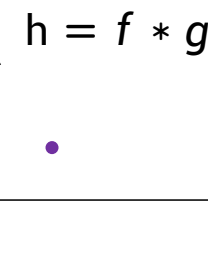
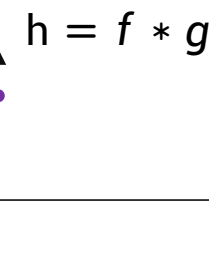
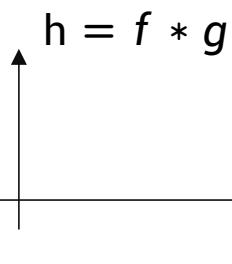
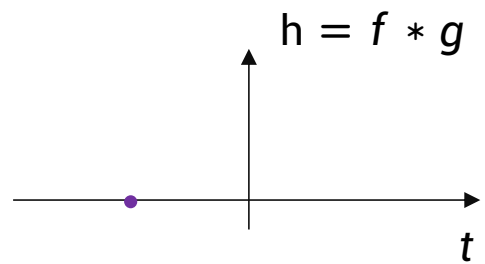
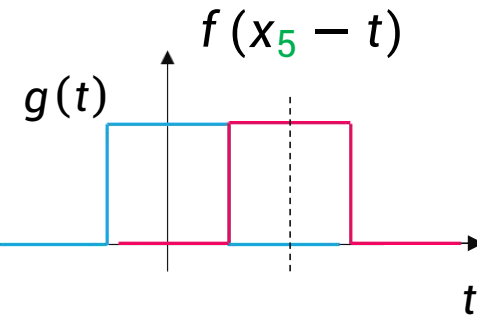
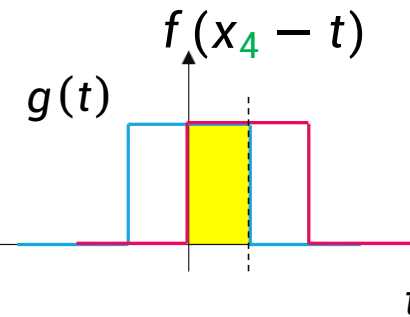
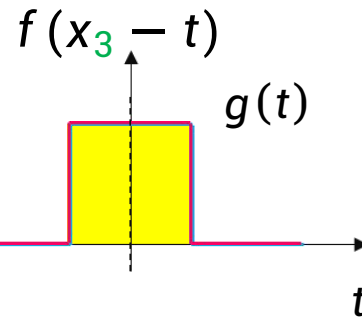
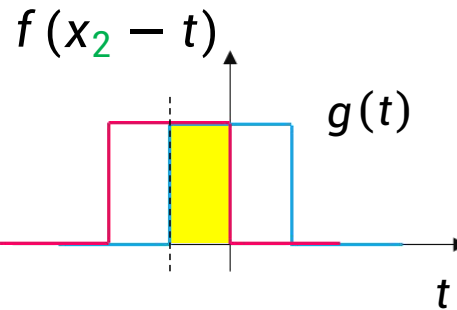
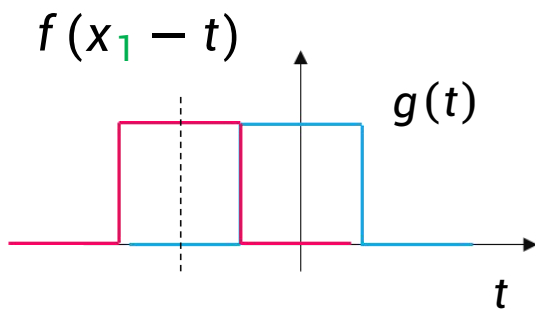
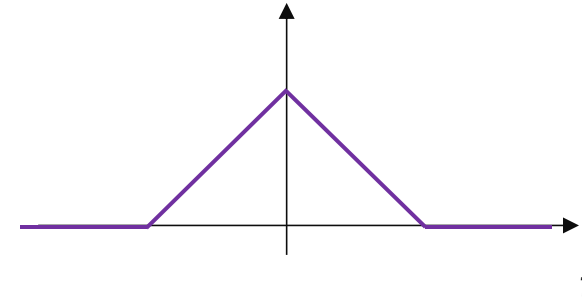
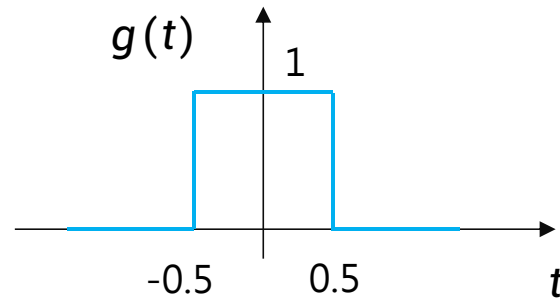
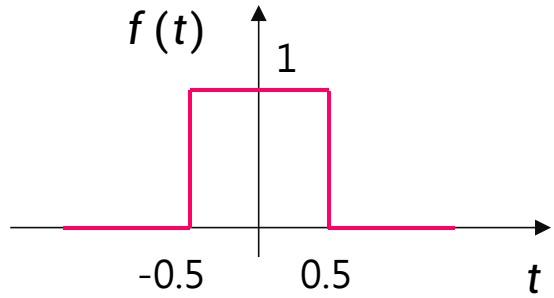


6.1 Convolution

Case 2

$$f(x) = g(x) = \begin{cases} 1 & \text{if } |x| \leq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

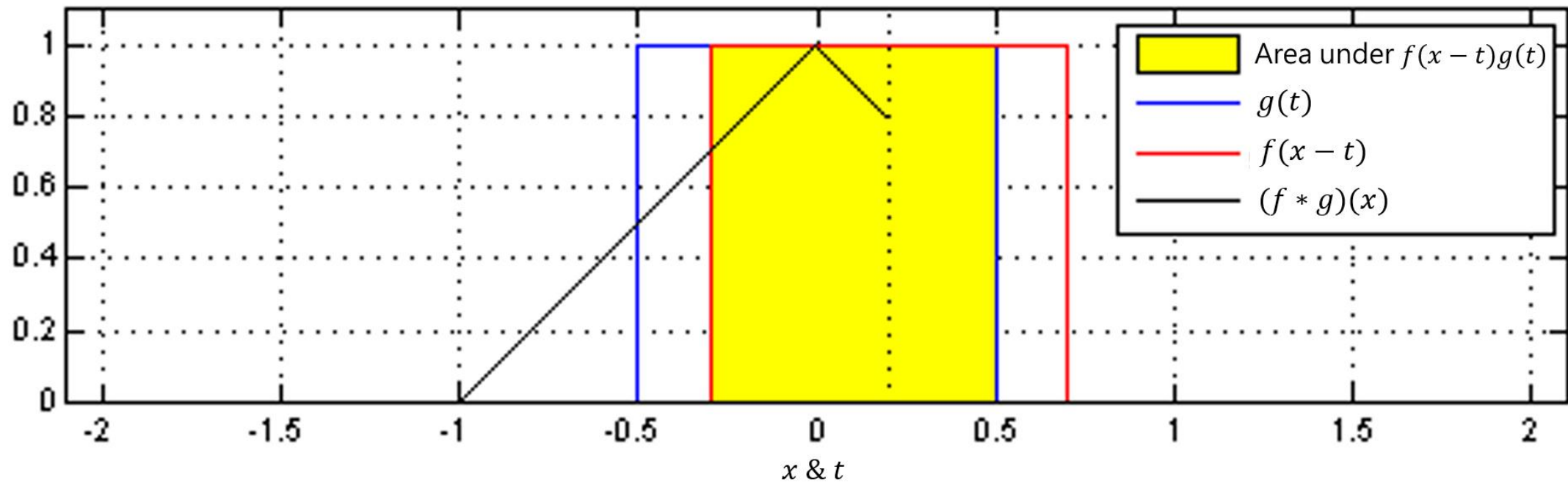
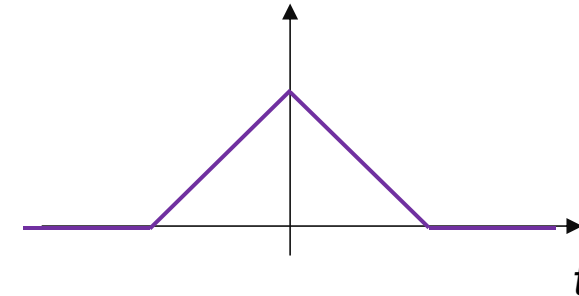
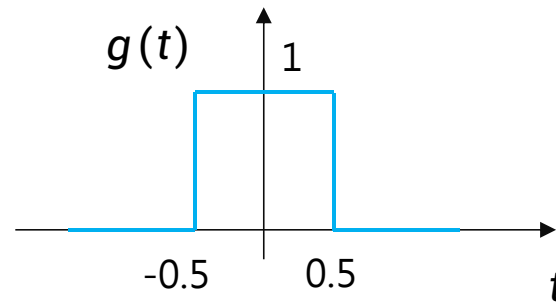
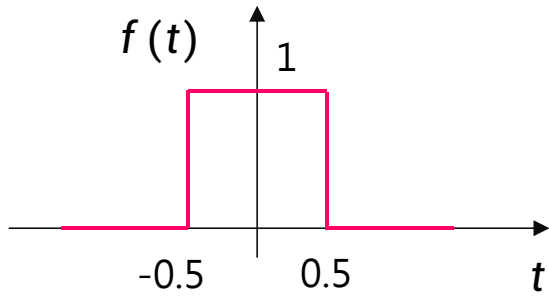
$$h = f * g = \int_{-\infty}^{\infty} f(x-t)g(t)dt$$



6.1 Convolution

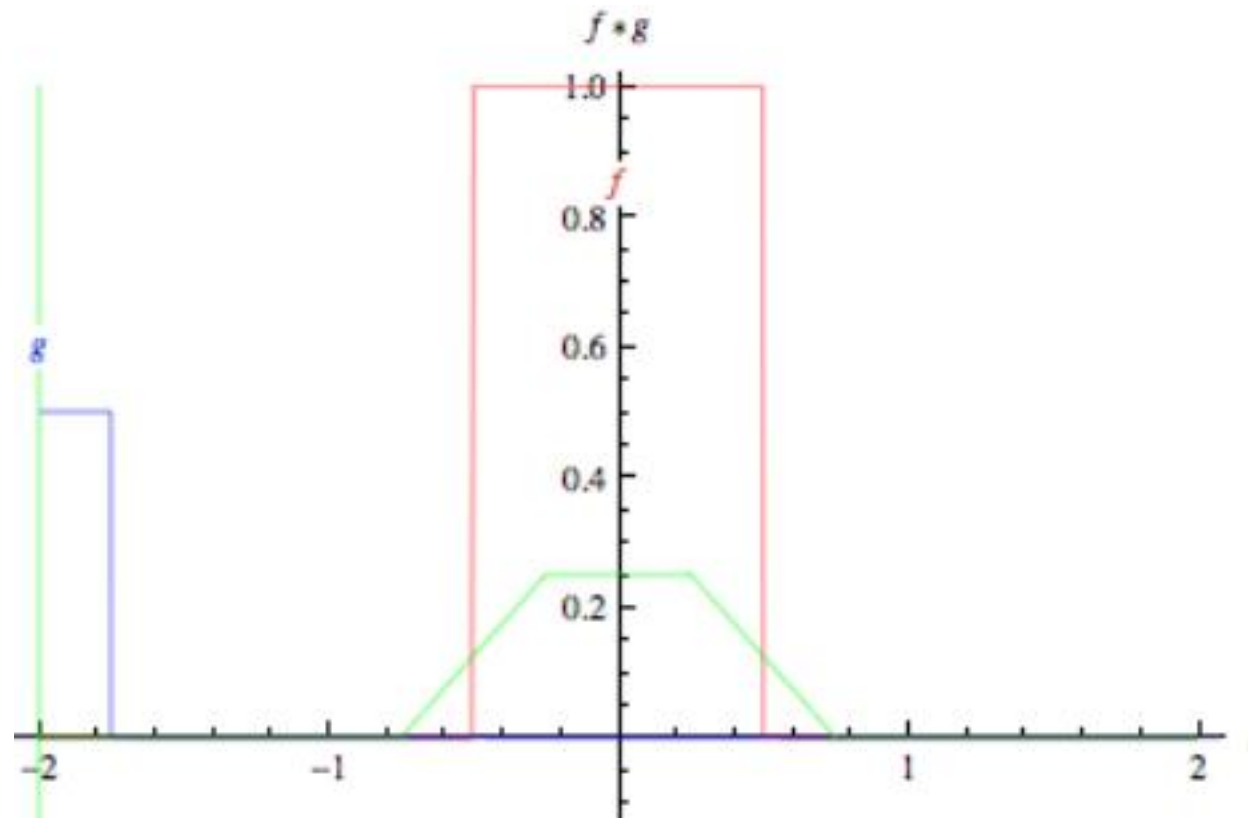
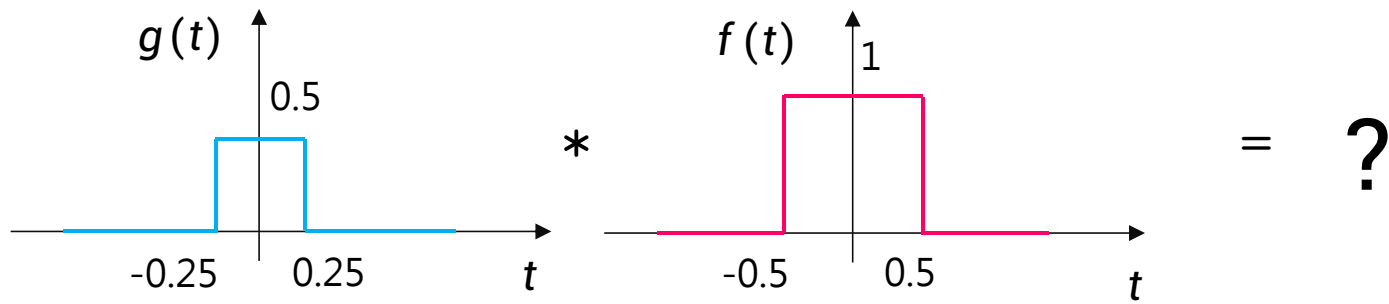
Case 2

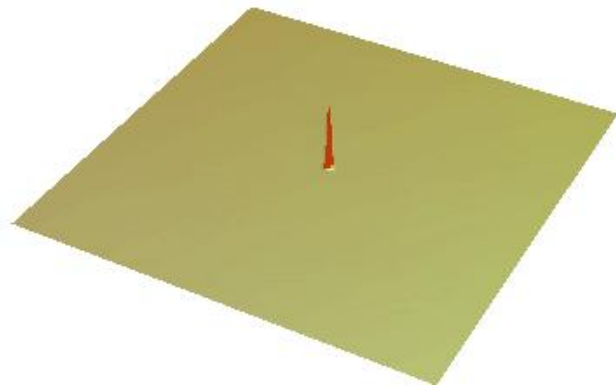
$$h = f * g = \int_{-\infty}^{\infty} f(x-t)g(t)dt$$



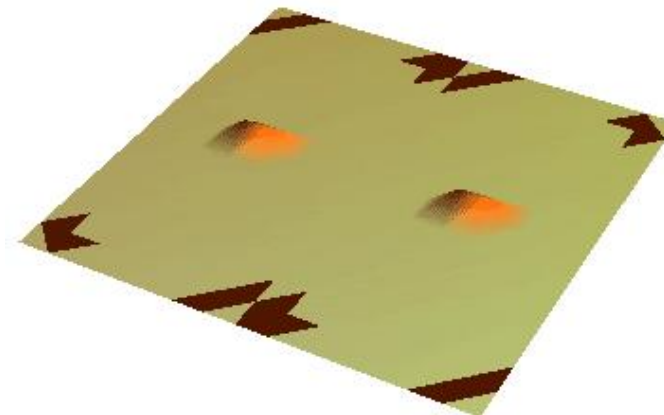
6.1 Convolution

Case 3





one stone wave



two stones wave

*Dirac Delta Function

Generalized Functions

$$g_k(t - a) = \begin{cases} \frac{1}{k} & \text{if } a \leq t \leq a + k \\ 0 & \text{otherwise} \end{cases}$$

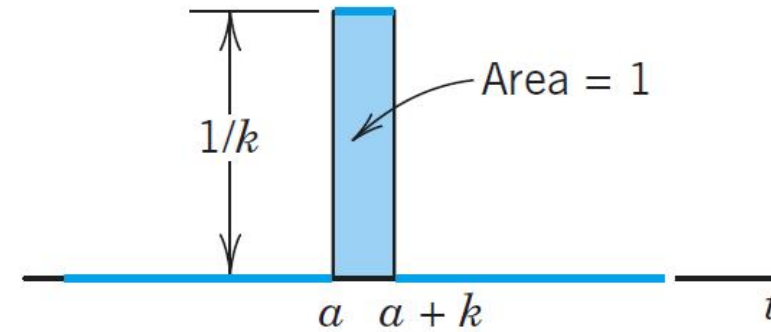
$$I_k = \int_{-\infty}^{\infty} g_k(t - a) dt = \int_a^{a+k} \frac{1}{k} dt = 1$$

When $k \rightarrow 0$ $\delta(t - a) = \lim_{k \rightarrow 0} g_k(t - a)$

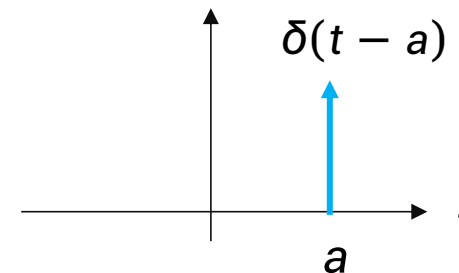
$$\delta(t - a) = \begin{cases} \infty & \text{if } t = a \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t - a) dt = 1$$

When $a = 0$

$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t) dt = 1$$



The function $g_k(t - a)$

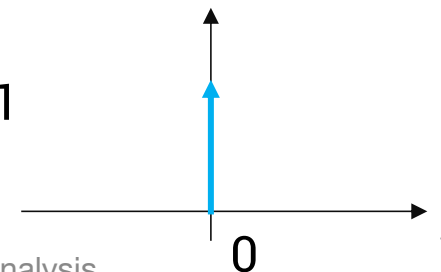


Theorem ($a = 0$)

$$\int_{-\infty}^{\infty} f(t) \delta(t) dt = f(0)$$

General form ($a \neq 0$)

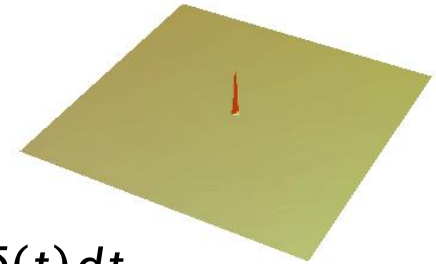
$$\int_{-\infty}^{\infty} f(t) \delta(t - a) dt = f(a)$$



6.1 Convolution

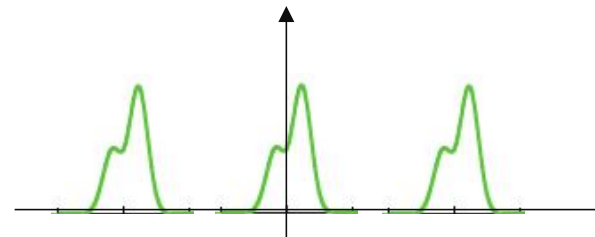
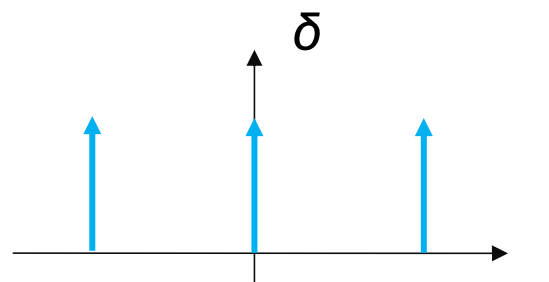
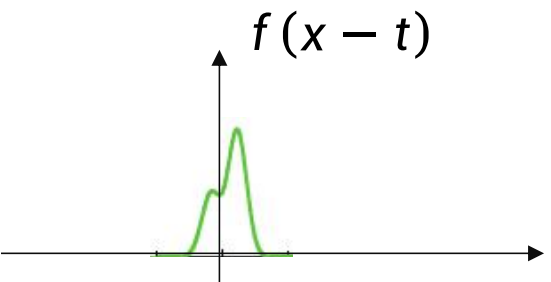
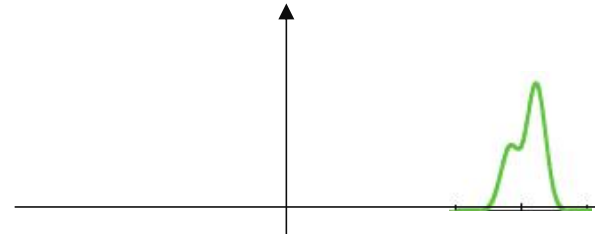
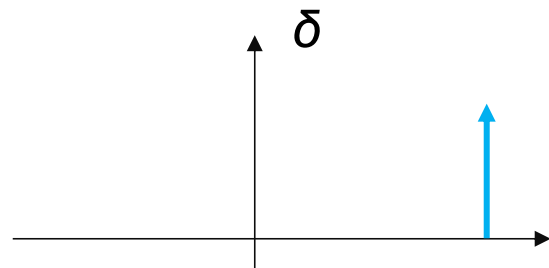
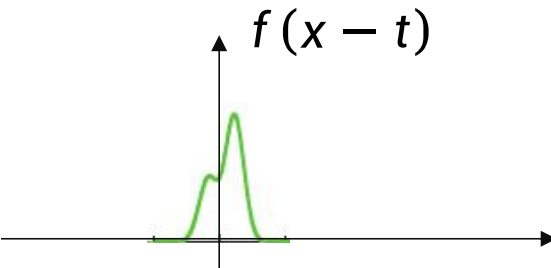
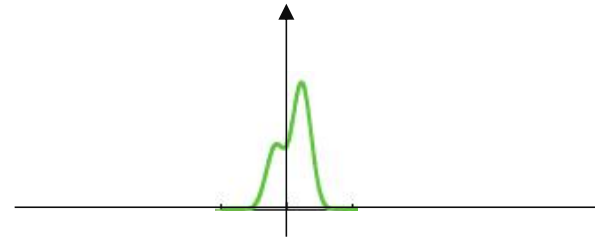
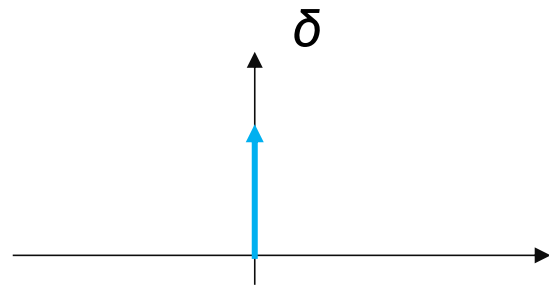
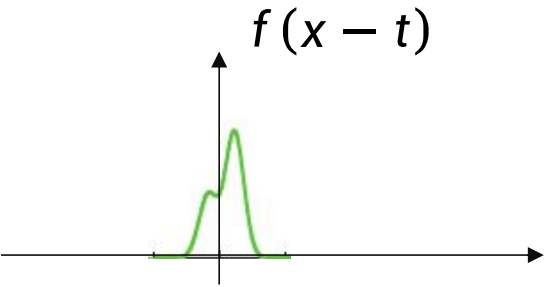
Case 4

$$h = f * \delta = \int_{-\infty}^{\infty} f(x-t)\delta(t)dt$$



$$h = f * \delta = \int_{-\infty}^{\infty} f(x-t)\delta(t)dt$$

$$h = f * \delta = \int_{-\infty}^{\infty} f(x-t)\delta(t)dt$$



6.1 Convolution

$$(f * g)(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x-t)g(t)dt \quad (6.1) \quad \text{Convolution}$$

From (6.1), by making a change of variables $t \leftrightarrow t - x$, we will see that **convolution is a commutative operation**, that is

$$(f * g)(x) = (g * f)(x)$$

6.1 Convolution

Example 6.1 Convolution with the cosine

Suppose that f is integrable and even (i.e. $f(-x) = f(x)$ for all x) and let $g(x) = \cos ax$.

Show that for all real numbers a : $(f * g)(x) = \cos(ax) \hat{f}(a)$.

Solution:

From (6.1) and the fact $f * g = g * f$, we have

$$\begin{aligned}(f * g)(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) \cos[a(x-t)] dt \\&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) [\cos(ax) \cos(at) + \sin(ax) \sin(at)] dt \\&= \frac{1}{\sqrt{2\pi}} \left[\cos(ax) \int_{-\infty}^{\infty} f(t) \cos(at) dt + \sin(ax) \int_{-\infty}^{\infty} f(t) \sin(at) dt \right]\end{aligned}$$

Since f is even, then the product $f(t) \sin(at)$ is odd,

thus by (2.4), $\int_{-\infty}^{\infty} f(t) \sin(at) dt = 0$, and so

$$\begin{aligned}(f * g)(x) &= \cos(ax) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) \cos(at) dt + 0 \\&= \cos(ax) \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^{\infty} f(t) \cos(at) dt - i \int_{-\infty}^{\infty} f(t) \sin(at) dt \right] \\&= \cos(ax) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} [f(t) \cos(at) - i f(t) \sin(at)] dt \\&= \cos(ax) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) [\cos(at) - i \sin(at)] dt \\&= \cos(ax) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-iat} dt \quad (\text{by Euler's formula}) \\&= \cos(ax) \hat{f}(a) \quad (\text{by (5.5)})\end{aligned}$$

6.1 Convolution

Example 6.2 Convolution as an average

For $n = 1, 2, \dots$, let

$$g_n(x) = \begin{cases} n\sqrt{\frac{\pi}{2}} & \text{if } -\frac{1}{n} < x < \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

Suppose that f is continuous on $(-\infty, \infty)$ and let F denote an antiderivative of f , i.e. $F' = f$.

Show that

$$(f * g_n)(x) = \frac{F\left(x + \frac{1}{n}\right) - F\left(x - \frac{1}{n}\right)}{\frac{2}{n}}$$

F : an antiderivative of f
 $F' = f$

Script $\mathcal{F}$: Fourier Transform
 $\hat{f}(\omega) = \mathcal{F}[f(x)]$

6.1 Convolution

Solution:

From (6.1), we have

$$(f * g_n)(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x-t) g_n(t) dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\frac{1}{n}}^{\frac{1}{n}} f(x-t) n \sqrt{\frac{\pi}{2}} dt$$

$$= \frac{n}{2} \int_{-\frac{1}{n}}^{\frac{1}{n}} f(x-t) dt$$

$$= \frac{n}{2} \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} f(t) dt \quad (\text{change } t \text{ to } x-t)$$

$$= \frac{n}{2} F(t) \Big|_{x-\frac{1}{n}}^{x+\frac{1}{n}}$$

$$= \frac{n}{2} \left[F\left(x+\frac{1}{n}\right) - F\left(x-\frac{1}{n}\right) \right]$$

When $n \rightarrow \infty$,

$$\begin{aligned} \lim_{n \rightarrow \infty} (f * g_n)(x) &= \lim_{n \rightarrow \infty} \frac{n}{2} \left[F\left(x+\frac{1}{n}\right) - F\left(x-\frac{1}{n}\right) \right] \\ &= \frac{1}{2} \lim_{\frac{1}{n} \rightarrow 0} \left[\frac{F\left(x+\frac{1}{n}\right) - F(x)}{\frac{1}{n}} + \frac{F\left(x+(-\frac{1}{n})\right) - F(x)}{(-\frac{1}{n})} \right] \end{aligned}$$

$$= F'(x) \quad (\text{by definition of derivative})$$

We know $F'(x) = f(x)$ by the fundamental theorem of calculus.

Then $\lim_{n \rightarrow \infty} (f * g_n)(x) = f(x)$.

6.1 Convolution

Theorem 6.1 Fourier Transforms of Convolutions (Convolution Theorem)

$$\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g) \quad (6.2)$$

$$(f * g)(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x-t)g(t)dt$$

Convolution in textbook

Theorem 6.1 is expressed by saying that the Fourier transform takes **convolution operation** into **product operation**.

Suppose that f and g are integrable; then

$$\mathcal{F}(f * g) = \sqrt{2\pi} \mathcal{F}(f)\mathcal{F}(g)$$

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-t)g(t)dt$$

Convolution in wikipedia

6.1 Convolution

Proof:

Using (6.1) and (5.3), we have

$$\mathcal{F}(f * g) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (f * g) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x-t) g(t) dt \right] e^{-i\omega x} dx$$

(by interchanging the order of integration)

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x-t) e^{-i\omega x} dx g(t) dt$$

(let $u = x - t$, $du = dx$)

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(u) e^{-i\omega(u+t)} du g(t) dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(u) e^{-i\omega u} du g(t) e^{-i\omega t} dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathcal{F}(f) g(t) e^{-i\omega t} dt$$

$$= \mathcal{F}(f) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(t) e^{-i\omega t} dt$$

$$= \mathcal{F}(f) \mathcal{F}(g)$$

6.1 Convolution

Example 6.3

Prove that convolution has the associative property: $f * (g * h) = (f * g) * h$

Proof:

Let $u = f * (g * h)$ and $v = (f * g) * h$. Then by Theorem 6.1

$$\begin{aligned}\mathcal{F}(u) &= \mathcal{F}(f * (g * h)) \\ &= \mathcal{F}(f) \cdot \mathcal{F}(g * h) \\ &= \mathcal{F}(f) \cdot (\mathcal{F}(g) \cdot \mathcal{F}(h)) \\ &= (\mathcal{F}(f) \cdot \mathcal{F}(g)) \cdot \mathcal{F}(h) \\ &= \mathcal{F}(f * g) \cdot \mathcal{F}(h) \\ &= \mathcal{F}((f * g) * h) \\ &= \mathcal{F}(v)\end{aligned}$$

Hence $u = v$ or $f * (g * h) = (f * g) * h$

6.1 Convolution

Example 6.4 Fourier transform of a convolution

Consider the function $f(x) = 1$ if $|x| < 1$ and 0 otherwise. The graph of this function is shown in [Figure 6.1](#). From [Example 5.3](#), we have

$$\hat{f}(\omega) = \sqrt{\frac{2}{\pi}} \frac{\sin \omega}{\omega}$$

We want to compute $f * f$, the convolution of f with itself.

Solution:

We do not directly compute from (6.1),

we will use Theorem 6.1 as follows.

$$\mathcal{F}(f * f) = \hat{f} \cdot \hat{f} = \hat{f}^2 = \left(\sqrt{\frac{2}{\pi}} \frac{\sin \omega}{\omega} \right)^2 = \frac{2}{\pi} \frac{\sin^2 \omega}{\omega^2}$$

To obtain $f * f$, we can use inverse Fourier transform.

If we check the table of Fourier transforms, we find

$$(f * f)(x) = \mathcal{F}^{-1} \left(\frac{2}{\pi} \frac{\sin^2 \omega}{\omega^2} \right) = \begin{cases} \sqrt{\frac{2}{\pi}} \left(1 - \frac{|x|}{2} \right) & \text{if } |x| < 2 \\ 0 & \text{if } |x| \geq 2. \end{cases}$$

The graph of $(f * f)(x)$ is shown in [Figure 6.2](#).

Note that $f * f$ is continuous even though f is not continuous.

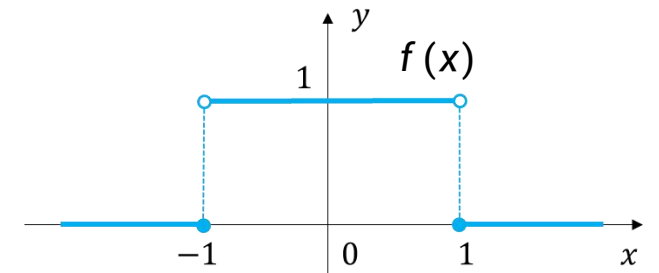


Figure 6.1 Graph of f .

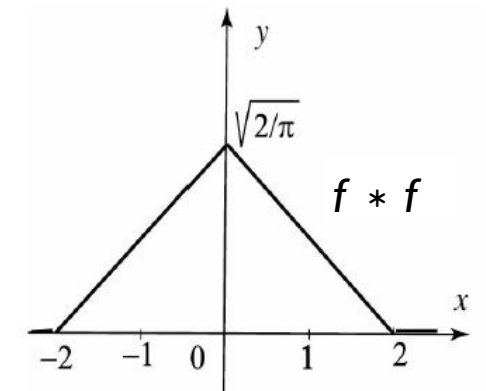


Figure 6.2 Graph of $f * f$.

6.1 Convolution

Table of Fourier Transform

$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{ix\omega} d\omega$		$\hat{f}(\omega) = \mathcal{F}(f)(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$
1.	$\begin{cases} 1 & \text{if } x < a \\ 0 & \text{if } x > a \end{cases}$	$\sqrt{\frac{2}{\pi}} \frac{\sin a\omega}{\omega}$
2.	$\begin{cases} 1 & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$	$\frac{i (e^{-ib\omega} - e^{ia\omega})}{\sqrt{2\pi}\omega}$
3.	$\begin{cases} 1 - \frac{ x }{a} & \text{if } x < a \\ 0 & \text{if } x > a \end{cases} \quad a > 0$	$2 \sqrt{\frac{2}{\pi}} \frac{\sin^2(\frac{a\omega}{2})}{a\omega^2}$
4.	$\begin{cases} x & \text{if } x < a \\ 0 & \text{if } x > a \end{cases} \quad a > 0$	$i \sqrt{\frac{2}{\pi}} \frac{a\omega \cos(a\omega) - \sin(a\omega)}{\omega^2}$

⋮

Review for Lecture 6

- Convolution
- Convolution Theorem
- *Dirac Delta Function

Assignment

Please Check <https://web-ext.u-aizu.ac.jp/~xiangli/teaching/MA05/index.html>

Reading materials: Lecture 6 Slides

References

[1] Nakhlé H. Asmar, *Partial Differential Equations with Fourier Series and Boundary Value Problems 2nd Edition*, 2004

[2] Wikipedia

4/27/2026